

CONVEXITYEDGE

Technical Documentation

Options Volatility Analytics Platform

Black-Scholes

FastAPI

SvelteKit

IV Surface

Greeks

Strategy Builder

VERSION

1.0.0

STACK

Python 3.14 · Node 22

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Project Overview

ConvexityEdge is a professional-grade, full-stack options volatility analytics terminal inspired by Bloomberg Terminal and ThinkorSwim. It provides real-time implied volatility analysis, options chain visualization, strategy building, and trade planning — all powered by Black-Scholes pricing.

1.1 Purpose & Goals

The platform is designed to give retail traders institutional-quality options analytics without a Bloomberg subscription. Every number displayed is computed from first principles using the Black-Scholes framework — no black-box vendor data for pricing.

GOAL	IMPLEMENTATION	STATUS
Real-time options chain with Greeks	yfinance + BS engine, 60s refresh	Done
IV Smile, Term Structure, 3D Surface	IV solver + ECharts / echarts-gl	Done
9-strategy payoff diagram builder	Analytical payoff + ECharts	Done
Trade analysis (Grid Reaper style)	Vol-floor scaling model	Done
Analytics dashboard	PCR, OI exposure, Greeks dashboard	Done
Auto dark/light mode by IST time	IST timezone store + Tailwind classes	Done
Full persistence layer	PostgreSQL + Redis (graceful degrade)	Done

1.2 Technology Stack

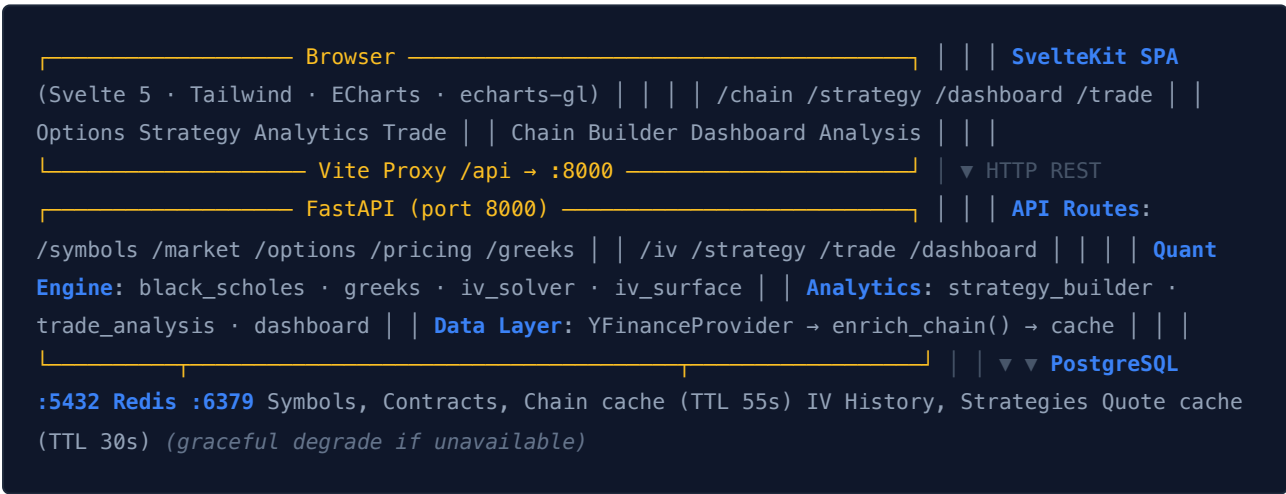
► Backend

► Frontend

TECHNOLOGY	ROLE
Python 3.14	Runtime
FastAPI 0.115	Async HTTP framework
NumPy / SciPy	Vectorized BS pricing
yfinance	Market data provider
SQLAlchemy async	ORM layer
asyncpg	PostgreSQL async driver
redis-py async	Cache layer
orjson	Fast JSON serialization
uvicorn	ASGI server

TECHNOLOGY	ROLE
Svelte 5	UI component framework
SvelteKit 2	App framework + routing
TypeScript	Type safety
Tailwind CSS 3	Utility-first styling
Apache ECharts 5	Charts (canvas)
echarts-gl	3D surface (WebGL)
Vite 6	Build tool / dev server

1.3 System Architecture



Mathematical Foundation

All pricing in ConvexityEdge is based purely on the Black-Scholes-Merton model for European options. No stochastic volatility (Heston), no stochastic interest rates (Vasicek/CIR). This section explains every formula used, from first principles.

2.1 Black-Scholes Model

Inputs

SYMBOL	NAME	TYPICAL VALUE	NOTES
S	Spot price	Market price	Current underlying price
K	Strike price	Option contract	The agreed exercise price
T	Time to expiry	DTE / 365	Expressed in years
r	Risk-free rate	~0.053 (5.3%)	US 10-year treasury yield
q	Dividend yield	~0.013 for SPY	Continuous dividend yield
σ	Volatility	0.15–0.80	Annualized implied volatility

Intermediate Values (d1 and d2)

$$d_1 = [\ln(S/K) + (r - q + \sigma^2/2) \cdot T] / (\sigma \cdot \sqrt{T})$$

$$d_2 = d_1 - \sigma \cdot \sqrt{T}$$

$\ln(S/K)$ is the **log-moneyness** — positive for ITM calls, negative for OTM calls. $\sigma \cdot \sqrt{T}$ is the expected one-standard-deviation move over the option's life. $N(d_2)$ is the risk-neutral probability that the option expires in-the-money.

Option Prices

$$\text{Call} = S \cdot e^{-qT} \cdot N(d_1) - K \cdot e^{-rT} \cdot N(d_2)$$

$$\text{Put} = K \cdot e^{-rT} \cdot N(-d_2) - S \cdot e^{-qT} \cdot N(-d_1)$$

$N(x)$ is the standard normal cumulative distribution function — the probability that a draw from $N(0,1)$ is $\leq x$. Implemented as `scipy.stats.norm.cdf(x)`. The exponential discount factors $e^{(-rT)}$ and $e^{(-qT)}$ account for the time value of money and foregone dividends.

PUT-CALL PARITY CHECK

$$C - P = S \cdot e^{-qT} - K \cdot e^{-rT}$$

This relationship must hold for any arbitrage-free market. The backend validates this for all computed prices.

► Vectorized Batch Pricing

For pricing an entire options chain (300+ contracts), the engine uses NumPy to compute all prices simultaneously rather than looping:

```
def bs_price_batch(S, K_arr, T_arr, r, q, sigma_arr, types):  
    # All arrays - NumPy vectorizes across all 300+ contracts at once  
    d1 = (np.log(S/K_arr) + (r - q + 0.5*sigma_arr**2)*T_arr) / (sigma_arr*np.sqrt(T_arr))  
    d2 = d1 - sigma_arr * np.sqrt(T_arr)  
    call_px = S*np.exp(-q*T_arr)*norm.cdf(d1) - K_arr*np.exp(-r*T_arr)*norm.cdf(d2)  
    put_px = K_arr*np.exp(-r*T_arr)*norm.cdf(-d2) - S*np.exp(-q*T_arr)*norm.cdf(-d1)  
    return np.where(types == 'C', call_px, put_px)
```

Performance: 10,000 contracts priced in ~4ms — well within the 2,000ms target.

2.2 The Greeks

Greeks measure the sensitivity of option price to changes in each input. All are computed analytically (exact closed-form derivatives), not numerically (bumping inputs). Analytical Greeks are faster, exact, and have no finite-difference error.

► Delta (Δ) — Price Sensitivity

$$\begin{aligned}\text{Call } \Delta &= e^{-qT} \cdot N(d_1) \in [0, 1] \\ \text{Put } \Delta &= -e^{-qT} \cdot N(-d_1) \in [-1, 0]\end{aligned}$$

Delta tells you: if the stock moves \$1, how much does the option price move? An ATM call has $\Delta \approx 0.50$. A deep ITM call has $\Delta \approx 1.0$ (moves dollar-for-dollar). Delta is also the hedge ratio — short 0.5 shares to delta-hedge one long ATM call.

► Gamma (Γ) — Delta Sensitivity

$$\Gamma = e^{-qT} \cdot \phi(d_1) / (S \cdot \sigma \cdot \sqrt{T}) \text{ [same for calls and puts]}$$

$\phi(x)$ is the standard normal PDF. Gamma peaks at-the-money and near expiry — this is where delta changes fastest and delta-hedging requires the most frequent rebalancing. High gamma = dangerous near expiry.

► Vega (v) — Volatility Sensitivity

$$v = S \cdot e^{-qT} \cdot \phi(d_1) \cdot \sqrt{T} / 100 \text{ [per 1\% change in IV, same for calls and puts]}$$

Vega is largest for ATM options and long-dated options. The $\div 100$ makes it readable: "option price changes by \$0.12 per 1% move in IV." Selling premium = short vega (you lose when IV spikes).

▸ Theta (Θ) — Time Decay

$$\theta_{\text{call}} = [-S \cdot e^{-qT} \cdot \phi(d_1) \cdot \sigma / (2\sqrt{T}) - r \cdot K \cdot e^{-rT} \cdot N(d_2) + q \cdot S \cdot e^{-qT} \cdot N(d_1)] / 365$$

Theta is always negative for long options — you lose time value every day. It accelerates as expiry approaches. Theta and Gamma are a trade-off: high gamma (good for long gamma scalpers) comes with high negative theta (you pay for it daily).

▸ Rho (ρ) — Interest Rate Sensitivity

$$\rho_{\text{call}} = K \cdot T \cdot e^{-rT} \cdot N(d_2) / 100$$

$$\rho_{\text{put}} = -K \cdot T \cdot e^{-rT} \cdot N(-d_2) / 100$$

Rho is the least important Greek for short-dated options. For long-dated LEAPS, a 1% rate change can meaningfully affect call prices.

2.3 Implied Volatility Solver

Implied volatility (IV) is the *reverse problem*: given an observed market price, what σ would Black-Scholes need to produce that price? There is no algebraic inverse — we must search numerically.

▸ Initial Guess (Brenner-Subrahmanyam Approximation)

$$\sigma_0 \approx \sqrt{(2\pi/T)} \cdot (C / S)$$

This approximation is accurate for ATM options and places the starting point close to the solution, reducing the number of iterations needed.

▸ Newton-Raphson Method (Primary)

$$\sigma_{\{n+1\}} = \sigma_n - (BS(\sigma_n) - \text{Market Price}) / \text{Vega}(\sigma_n)$$

Newton's method converges quadratically near the root — typically 3–5 iterations. It fails when Vega is near zero (deep ITM or OTM options where the price is nearly all intrinsic value).


```
def solve_iv_newton(market_price, S, K, T, r, q, option_type, max_iter=50):
    sigma = sqrt(2*pi/T) * (market_price/S) # Brenner-Subrahmanyam guess
    for _ in range(max_iter):
        price = bs_price(S, K, T, r, q, sigma, option_type)
        vega = bs_vega(S, K, T, r, q, sigma)
        diff = price - market_price
        if abs(diff) < 1e-8: return sigma # converged
        if abs(vega) < 1e-10: raise ValueError("vega too small")
        sigma -= diff / vega
        sigma = max(0.001, min(sigma, 10.0)) # clamp to [0.1%, 1000%]
    raise ValueError("did not converge")
```

▸ Brent's Method (Fallback)

Brent's method is a hybrid of bisection, secant, and inverse quadratic interpolation. It guarantees convergence as long as a bracket exists (i.e., the function changes sign over `[σlo, σhi]`). We bracket `[0.001, 10.0]`. It is slower than Newton but never fails.

IV SOLVER STRATEGY

Primary: Newton-Raphson (fast, 3–5 iters) → **Fallback:** Brent's method (slow, guaranteed). The solver chain automatically falls back on any exception (zero vega, non-convergence, numerical overflow).

2.4 IV Surface Construction

▸ IV Smile

Plot IV vs. strike for a single expiry. Calls and puts are plotted separately. In real markets, the smile is not flat — deep OTM puts are more expensive than calls (negative skew) because of crash fear.

▸ IV Term Structure

Plot ATM IV vs. DTE across all available expirations. Normal markets show upward-sloping term structure (more uncertainty further out). Inverted term structure (front month IV > back month) signals a near-term event risk (earnings, FOMC).

▸ IV Rank

$$\text{IV Rank} = \left(\text{Current IV} - 52\text{-week Low} \right) / \left(52\text{-week High} - 52\text{-week Low} \right) \times 100$$

A rank of 80 means IV is in the 80th percentile of its past year. High IV Rank is favourable for selling premium (straddles, iron condors). Low IV Rank favours buying options.

▸ 3D IV Surface

A 2D grid of (strike, DTE) with IV as the Z-axis. Built by running the IV solver for every contract in the chain, then rendering with echarts-gl's WebGL surface series. The colour map goes from deep blue (low IV) through yellow to red (high IV).

Backend Architecture

The backend is a Python FastAPI application with a fully async architecture. It separates concerns into a quant engine (pure math), analytics layer, data pipeline, and API layer.

3.1 Project Structure

```

backend/
├── app/
│   ├── main.py          # FastAPI app, lifespan, CORS, ORJSON
│   ├── core/
│   │   ├── config.py    # Pydantic settings (env vars)
│   │   ├── database.py  # SQLAlchemy async engine
│   │   └── redis.py      # Redis client + graceful degrade helpers
│   ├── api/v1/routes/
│   │   ├── symbols.py   # GET /symbols/search
│   │   ├── market.py    # GET /market/quote, /market/chart
│   │   ├── options.py   # GET /options/expiries, /options/chain
│   │   ├── pricing.py   # POST /pricing/black-scholes
│   │   ├── greeks.py    # POST /greeks/calculate
│   │   ├── iv.py        # GET /iv/smile, /iv/term-structure, /iv/surface
│   │   ├── strategy.py  # POST /strategy/build, /strategy/payoff
│   │   ├── trade.py     # POST /trade/analyze
│   │   └── dashboard.py # GET /dashboard/analytics/{symbol}
│   └── models/          # SQLAlchemy ORM models
├── quant_engine/
│   ├── black_scholes.py # bs_price, bs_price_batch
│   ├── greeks.py        # calculate_greeks (vectorized)
│   ├── iv_solver.py     # Newton-Raphson + Brent
│   └── iv_surface.py     # smile, term structure, 3D surface
├── analytics/
│   ├── strategy_builder.py # 9 strategy templates + payoff
│   ├── trade_analysis.py  # vol-floor scaling model
│   └── dashboard_analytics.py
├── data/
│   ├── providers/
│   │   ├── base.py      # Abstract MarketDataProvider
│   │   └── yfinance_provider.py
│   └── pipeline.py       # enrich_chain, run_refresh_loop

```

3.2 FastAPI Application

▸ Lifespan Context Manager

FastAPI's `lifespan` replaces the deprecated `@app.on_event` startup/shutdown hooks. It uses an `async with` context manager — code before `yield` runs at startup, after `yield` at shutdown.

```
@asynccontextmanager
async def lifespan(app: FastAPI):
    logger.info("ConvexityEdge backend starting...")
    asyncio.create_task(run_refresh_loop()) # start 60s background refresh
    try:
        await get_redis() # warm up Redis connection
    except Exception:
        logger.warning("Redis unavailable - running without cache")
    yield # app is now serving requests
    # shutdown cleanup happens here
```

► ORJSON Responses

All endpoints use `ORJSONResponse` instead of the default JSON response. ORJSON is 2–10x faster than Python's stdlib `json` module and natively handles NumPy arrays, datetime objects, and Decimal types.

3.3 Quant Engine

► black_scholes.py

FUNCTION	INPUTS	OUTPUT	NOTES
<code>_d1_d2()</code>	S, K, T, r, q, σ	(d1, d2)	Private, used internally
<code>bs_price()</code>	S, K, T, r, q, σ , type	float	Single option
<code>bs_price_with_decomposition()</code>	same	dict	Price + intrinsic + time value
<code>bs_price_batch()</code>	S, K_arr, T_arr, σ _arr	np.ndarray	Vectorized, entire chain

► greeks.py

FUNCTION	DESCRIPTION
<code>calculate_greeks(S, K, T, r, q, σ, type)</code>	Returns full Greeks dict for one option
<code>greeks_scalar()</code>	Same, returns Pydantic model
<code>greeks_batch(S, K_arr, T_arr, σ_arr, types)</code>	Vectorized — all Greeks for entire chain

► iv_solver.py

```
# Solver chain: Newton-Raphson first, Brent as fallback
def solve_iv(market_price, S, K, T, r, q, option_type) -> float | None:
    if market_price <= 0 or T <= 0:
        return None
    try:
        return solve_iv_newton(market_price, S, K, T, r, q, option_type)
    except Exception:
        try:
            return _solve_brent(market_price, S, K, T, r, q, option_type)
        except Exception:
            return None # deep ITM/OTM with no market price
```

3.4 Data Pipeline

▸ Market Data Provider (Abstract Pattern)

The `MarketDataProvider` abstract base class defines the contract for all data sources. `YFinanceProvider` implements it using yfinance. Switching to a live data source (Polygon.io, Alpaca, IBKR) only requires writing a new provider class.

```
class MarketDataProvider(ABC):
    @abstractmethod
    async def get_quote(self, symbol: str) -> QuoteData: ...
    @abstractmethod
    async def get_option_chain(self, symbol: str, expiry: date) -> list[OptionContractData]: ...
```

▸ yfinance Thread Pool Pattern

yfinance makes synchronous HTTP requests. Running these directly in FastAPI's async event loop would *block* it, preventing other requests from being served. The solution is to run yfinance in a thread pool:

```
async def get_option_chain(self, symbol, expiry):
    def _fetch(): # synchronous – runs in thread
        ticker = yf.Ticker(symbol)
        chain = ticker.option_chain(expiry.isoformat())
        # parse calls and puts ...
    loop = asyncio.get_event_loop()
    return await loop.run_in_executor(None, _fetch) # non-blocking
```

▸ NaN-Safe Field Parsing

yfinance often returns `float('nan')` for missing fields (no trades, zero volume). `nan` is truthy in Python, so `nan or 0` returns `nan`, and `int(nan)` raises `ValueError`. The `_fi()` / `_ff()` helpers handle this:

```
def _fi(val, default=0) -> int:
    try:
        v = float(val)
        return default if (math.isnan(v) or math.isinf(v)) else int(v)
    except (TypeError, ValueError):
        return default
```

► Chain Enrichment

After fetching raw contract data from yfinance, `enrich_chain()` runs the full quant pipeline:

1. Solve IV for each contract using market bid/ask midpoint
2. Compute all five Greeks analytically
3. Mark ATM strike (closest to current spot price)
4. Compute chain statistics: ATM IV, HV-30d, PCR, OI exposure
5. Persist to PostgreSQL (if available)
6. Cache result in Redis with 55-second TTL

► Background Refresh Loop

```
async def run_refresh_loop():
    while True:
        await asyncio.sleep(60)
        for symbol in WATCHED_SYMBOLS:
            await refresh_symbol(symbol)  # re-fetch + re-enrich + re-cache
```

3.5 Strategy Builder

Long Call

+1 call @ ATM

Bullish. Unlimited upside, premium is max loss. Ideal when expecting a large up move.

Long Put

+1 put @ ATM

Bearish. Unlimited downside capture, premium is max loss. Hedge against long stock.

Bull Call Spread

+1 call ATM, -1 call OTM

Defined-risk bullish. Reduces cost vs naked call, caps upside at short strike.

Bear Put Spread

+1 put ATM, -1 put OTM

Defined-risk bearish. Cheaper than naked put, profit capped at short strike.

Straddle

+1 call ATM, +1 put ATM

Long volatility. Profits if stock moves significantly in either direction. Theta-heavy.

Strangle

+1 call OTM, +1 put OTM

Cheaper than straddle. Requires larger move to profit. Good before major events.

Iron Condor

-1 put OTM, +1 put far OTM
-1 call OTM, +1 call far OTM

Short volatility, range-bound. Max profit if stock stays between short strikes.

Butterfly

+1 call low, -2 call mid, +1 call high

Peaked payoff. Maximum profit if stock closes exactly at middle strike at expiry.

► Payoff Calculation

```
def calculate_payoff(legs, price_range):
    prices = np.linspace(price_range[0], price_range[1], 200)
    payoffs = np.zeros(len(prices))
    for leg in legs:
        intrinsic = np.maximum(0,
                                (prices - leg.strike) if leg.option_type == 'C'
                                else (leg.strike - prices))
        sign = 1 if leg.action == 'BUY' else -1
        payoffs += sign * leg.quantity * intrinsic - sign * leg.premium * 100
    return [PayoffPoint(price=p, pnl=v) for p, v in zip(prices, payoffs)]
```

3.6 Trade Analysis (Vol-Floor Model)

Inspired by Grid Reaper / ScaleOrder scaling strategies. Uses implied volatility to define a statistically-derived support zone for scaling into positions.

$$\text{Floor Value} = \text{Entry Price} \times (1 - IV \times \sqrt{\text{Period}})$$

Example: SPY=\$500, IV=20%, Monthly period:

$$\text{Floor} = 500 \times (1 - 0.20 \times \sqrt{(1/12)}) = 500 \times 0.9423 = \$471.15$$

This is a 1-standard-deviation downside move for the period. The grid levels are spaced equally between entry and floor, with capital allocated across levels to achieve a target profit if price recovers to entry.

3.7 Database & Caching

► PostgreSQL Schema (7 tables)

TABLE	PURPOSE	KEY COLUMNS
<code>symbols</code>	Tracked securities	ticker, name, exchange, asset_type
<code>option_contracts</code>	Option metadata	symbol, strike, expiry, option_type
<code>option_chain_snapshots</code>	Point-in-time chain data	contract_id, bid, ask, iv, greeks, timestamp
<code>iv_history</code>	IV time series for IV Rank	symbol, expiry, atm_iv, timestamp
<code>strategy_definitions</code>	9 built-in strategies	name, description, leg_templates
<code>strategy_legs</code>	User-saved strategies	strategy_id, option_type, action, quantity
<code>trade_analysis_results</code>	Saved trade plans	symbol, entry, floor, levels, timestamp

▸ Async SQLAlchemy + asyncpg

`psycopg2-binary` has no Python 3.14 wheel yet. `asyncpg` is a pure-Python async PostgreSQL driver — faster than `psycopg2` on async workloads and works on Python 3.14. The SQLAlchemy engine is configured with `pool_size=20` to handle concurrent requests.

▸ Redis Graceful Degradation

All Redis operations are wrapped in `try/except` that silently return `None` on any error. The endpoint re-fetches from yfinance on a cache miss. The app works identically with or without Redis — Redis is a performance optimization, not a dependency.

3.8 API Reference

▸ Market Data

GET	<code>/api/v1/market/quote/{symbol}</code>	Current price, change, volume, market cap
GET	<code>/api/v1/market/chart/{symbol}?period=3mo&interval=1d</code>	OHLCV bars for candlestick chart
GET	<code>/api/v1/options/expiries/{symbol}</code>	List of available expiry dates with DTE labels
GET	<code>/api/v1/options/chain/{symbol}?expiry=YYYY-MM-DD&strikes=30</code>	Full enriched chain: Greeks, IV, bid/ask for N strikes either side of ATM

▸ Pricing & Greeks

POST	<code>/api/v1/pricing/black-scholes</code>	Body: {S, K, T, r, q, sigma, option_type} → price + decomposition
POST	<code>/api/v1/greeks/calculate</code>	Body: same as above → {delta, gamma, vega, theta, rho}

► Implied Volatility

GET /api/v1/iv/smile/{symbol}?expiry=YYYY-MM-DD IV smile data points (calls + puts) for a single expiry

GET /api/v1/iv/term-structure/{symbol} ATM IV across all available expirations

GET /api/v1/iv/surface/{symbol} Full 2D grid of (strike, DTE, IV%) for 3D surface

GET /api/v1/iv/skew/{symbol}?expiry=YYYY-MM-DD Skew: call IV – put IV by moneyness

► Strategy & Trade

GET /api/v1/strategy/templates List of 9 strategy templates with descriptions

POST /api/v1/strategy/build Body: {strategy_name, symbol, expiry, underlying_price} → leg list

POST /api/v1/strategy/payoff Body: {legs, price_range?} → payoff curve, break-evens, max P&L

POST /api/v1/trade/analyze Body: trade params → floor, grid levels, expected move, risk/reward

GET /api/v1/dashboard/analytics/{symbol}?expiry=YYYY-MM-DD Full analytics: IV rank, HV, PCR, OI exposure, top volume strikes

Frontend Architecture

The frontend is a SvelteKit SPA running entirely client-side (`ssr=false`). It uses Svelte 5's reactive store system, Apache ECharts for all charts, and Tailwind CSS for a terminal-style dark theme.

4.1 SvelteKit & Project Structure

```
frontend/src/
├── routes/
│   ├── +layout.ts           # export const ssr = false (global CSR)
│   ├── +layout.svelte       # Navbar + theme sync
│   ├── +page.svelte         # Redirect → /chain
│   ├── chain/+page.svelte   # Options chain terminal
│   ├── strategy/+page.svelte # Strategy builder + payoff diagram
│   ├── dashboard/+page.svelte # Analytics dashboard
│   └── trade/+page.svelte   # Trade analysis panel
├── lib/
│   ├── api/
│   │   ├── client.ts       # Typed fetch wrappers for all endpoints
│   │   └── types.ts        # All TypeScript interfaces
│   ├── stores/
│   │   ├── market.ts       # selectedSymbol, optionChain, quote, etc.
│   │   ├── refresh.ts      # 60s refresh timer + countdown
│   │   └── theme.ts        # IST-aware dark/light theme store
│   ├── components/
│   │   ├── shared/         # Navbar, SymbolSearch, ExpirySelector
│   │   ├── OptionsChain/   # OptionsChainTable, ChainRow
│   │   ├── MarketChart/   # CandlestickChart
│   │   ├── IVCharts/       # IVSmile, IVTermStructure, IVSurface
│   │   ├── StrategyBuilder/ # StrategySelector, LegsTable, PayoffDiagram
│   │   ├── TradeAnalysis/  # TradePanel
│   │   └── Dashboard/      # MetricCard
│   ├── utils/
│   │   ├── formatting.ts   # fmtPrice, fmtPct, fmtVolume, signClass
│   │   └── colors.ts       # CHART_COLORS constant object
│   └── types/
│       └── echarts-gl.d.ts  # declare module 'echarts-gl'
```

► SSR Disabled Globally

SvelteKit defaults to server-side rendering (SSR), which pre-renders page HTML on the server before sending it to the browser. Our pages make `fetch('/api/...')` calls with relative URLs. On the server, there is no host to resolve relative URLs against, causing a crash. Setting `ssr = false` in the root `+layout.ts` turns off SSR for the entire app:

```
// src/routes/+layout.ts
export const ssr = false; // all pages render client-side only
```

4.2 Reactive Stores

Svelte stores are observable state containers. Any component can subscribe to them; when the value changes, all subscribers re-render automatically. The `$` prefix is syntactic sugar for `.subscribe()`.

```
// src/lib/stores/market.ts
export const selectedSymbol = writable<string>('SPY');
export const optionChain = writable<OptionChain | null>(null);

// derived store – recalculates when optionChain or quote changes
export const atmStrike = derived(
  [optionChain, quote],
  ([chain, $quote]) => {
    if (!$chain || !$quote) return null;
    return $chain.strikes
      .map(s => s.strike)
      .reduce((prev, curr) =>
        Math.abs(curr - $quote.price) < Math.abs(prev - $quote.price) ? curr : prev
      );
  }
);
```

In a component, using `$selectedSymbol` automatically subscribes and triggers re-render on change. No manual subscription management needed.

► Refresh Timer Store

The `refresh.ts` store manages two intervals: one that counts down from 60 to 0 every second (for the navbar ring), and one that fires every 60 seconds to trigger a data reload. Both are guarded with `if (!browser) return` so they never run during SSR.

4.3 API Client Layer

All backend calls go through typed wrapper functions in `client.ts`. They use the browser's native `fetch` API, and Vite proxies `/api` requests to `localhost:8000` in development.

```
// src/lib/api/client.ts
const BASE = '/api/v1';

async function request<T>(path: string, options?: RequestInit): Promise<T> {
  const res = await fetch(`${BASE}${path}`, {
    headers: { 'Content-Type': 'application/json' },
    ...options,
  });
  if (!res.ok) {
    const err = await res.json().catch(() => ({ detail: res.statusText }));
    throw new Error(err.detail ?? `HTTP ${res.status}`);
  }
  return res.json();
}

export const optionsApi = {
  chain: (symbol: string, expiry: string, strikes = 30) =>
    request<OptionChain>(`/options/chain/${symbol}?expiry=${expiry}&strikes=${strikes}`),
};
```

4.4 Pages & Components

▶ Chain Page (/chain)

The main options chain terminal. Layout: stat ticker bar at top → toolbar (expiry selector, strike count) → table (calls | strike | puts) → candlestick chart below.

- **Stat bar:** Underlying price, change%, ATM IV, HV-30d, PCR volume/OI, call/put volume
- **Chain table:** 8 columns each side (Vol, OI, Δ , Γ , v , Θ , Bid, Ask). ATM row highlighted in accent blue.
- **Greeks color coding:** Delta (blue), Gamma (green), Vega (purple), Theta (red)
- **Auto-refresh:** 60-second timer started in `onMount`, stopped in `onDestroy`

▶ Strategy Page (/strategy)

Left panel: strategy selector dropdown + legs table. Right panel: payoff diagram (ECharts line chart with area fill).

- Metric cards: Net Premium, Max Profit, Max Loss, Risk/Reward, Expected Move, Break-Evens
- Payoff diagram: green area above zero P&L, red area below. Blue dashed line at spot price, amber dashed lines at break-evens.

▶ Dashboard Page (/dashboard)

Analytics overview: 8 metric cards (ATM IV, IV Rank, HV-30d, IV-HV Spread, PCR, Total Volume, Greeks Exposure) + 3-panel chart grid (IV Smile, IV Term Structure, IV Surface 3D) + tables for top-volume calls/puts.

▶ Trade Page (/trade)

Left: TradePanel (Grid Reaper-style inputs — entry price, IV, expiry, preset levels, initial size, subsequent size, vol floor period). Right: Candlestick chart + IV Term Structure below.

► SymbolSearch Component

Debounced search (250ms) with dropdown results. Calls `GET /api/v1/symbols/search?q=`. On selection, updates the `selectedSymbol` store — this triggers reactive reloads in every page that watches `$selectedSymbol`.

► ChainRow Component

Renders one row of the options chain table. Receives the `ChainStrike` object (has `.call` and `.put` sub-objects). ATM rows get the `row-atm` CSS class for the blue highlight. Greeks are color-coded with inline style or Tailwind utility classes.

4.5 Charts (ECharts)

All charts follow the same pattern: `onMount` initializes the ECharts instance (canvas or WebGL), sets the option object, and a `ResizeObserver` handles responsive resizing. Reactive statements re-render when props change.

SVELTE 5 ONMOUNT PATTERN

In Svelte 5, `onMount(async () => { return cleanup })` does NOT call the `cleanup` function because Svelte ignores `Promise<cleanup>`. All chart components use a synchronous `onMount` that launches async work via an IIFE and returns `cleanup` synchronously.

```
// Correct pattern for all chart components
onMount(() => {
  let ro: ResizeObserver;
  (async () => {
    const echarts = await import('echarts'); // dynamic import
    chart = echarts.init(chartEl, null, { renderer: 'canvas' });
    ro = new ResizeObserver(() => chart.resize());
    ro.observe(chartEl);
    await loadData();
  })();
  return () => { if (ro) ro.disconnect(); if (chart) chart.dispose(); };
});
```

COMPONENT	CHART TYPE	RENDERER	KEY FEATURES
CandlestickChart	Candlestick + bar	Canvas	MA20, MA50, volume panel, dataZoom slider
IVSmile	Line (2 series)	Canvas	Calls (green) + Puts (red), smooth curve
IVTermStructure	Area line	Canvas	DTE on x-axis, ATM IV labels on points
IVSurface	3D surface mesh	WebGL	echarts-gl, colour-mapped by IV%
PayoffDiagram	Area line	Canvas	Mark lines: spot (blue), break-evens (amber)

▸ IVSurface — WebGL Renderer

The 3D surface uses `echarts-gl` which extends ECharts with WebGL rendering. The renderer is `'webgl'` (not `'canvas'`). Since ECharts' TypeScript types don't include `'webgl'` (it's added by the extension), a cast is needed: `renderer: 'webgl' as any`. The surface data is a flat array of `[strikeIndex, dteIndex, iv_pct]` tuples.

4.6 Theme System

The theme automatically switches based on Indian Standard Time (IST):

- **Light mode:** 09:00 – 15:30 IST (Indian market hours)
- **Dark mode:** All other times (pre-market, after-market, weekends)
- **User override:** Stored in `localStorage` as `ce_theme`. If present, auto-switching is disabled.

```
// src/lib/stores/theme.ts
function getISTHour(): number {
  const now = new Date();
  const utcMs = now.getTime() + now.getTimezoneOffset() * 60_000;
  const istMs = utcMs + 5.5 * 3_600_000; // UTC+5:30
  return new Date(istMs).getHours() + new Date(istMs).getMinutes() / 60;
}

function inferThemeFromIST(): Theme {
  const h = getISTHour();
  return (h >= 9 && h < 15.5) ? 'light' : 'dark';
}
```

The theme store calls `document.documentElement.classList.add('dark')` or `remove('dark')` which triggers Tailwind's dark-mode class strategy. Every minute, the layout's `setInterval` calls `theme.syncToIST()` to re-check.

Infrastructure & Deployment

5.1 Docker Compose

```
# docker/docker-compose.yml – four services
services:
  postgres:
    image: postgres:16-alpine
    environment: { POSTGRES_DB: convexityedge, POSTGRES_PASSWORD: ... }
    volumes: [pgdata:/var/lib/postgresql/data]

  redis:
    image: redis:7-alpine
    command: redis-server --maxmemory 256mb --maxmemory-policy allkeys-lru

  backend:
    build: ./backend
    depends_on: [postgres, redis]
    environment: { DATABASE_URL: ..., REDIS_URL: ... }
    ports: ["8000:8000"]

  frontend:
    build: ./frontend
    depends_on: [backend]
    ports: ["3000:3000"]
```

DEVELOPMENT VS PRODUCTION

In development, run the backend directly with `uvicorn app.main:app --reload` from the `backend/` directory, and the frontend with `npm run dev` from `frontend/`. Vite's dev proxy handles `/api → :8000`. For production, use Docker Compose.

5.2 Database Schema

```
-- database/schema.sql (key tables)
CREATE TABLE symbols (
    id SERIAL PRIMARY KEY,
    ticker VARCHAR(10) UNIQUE NOT NULL,
    name TEXT,
    exchange VARCHAR(20),
    asset_type VARCHAR(20) DEFAULT 'equity'
);

CREATE TABLE option_chain_snapshots (
    id BIGSERIAL PRIMARY KEY,
    contract_id INTEGER REFERENCES option_contracts(id),
    bid NUMERIC(12,4),
    ask NUMERIC(12,4),
    last NUMERIC(12,4),
    volume INTEGER,
    open_interest INTEGER,
    implied_vol NUMERIC(8,6),
    delta NUMERIC(8,6),
    gamma NUMERIC(8,6),
    vega NUMERIC(8,6),
    theta NUMERIC(8,6),
    rho NUMERIC(8,6),
    fetched_at TIMESTAMPTZ DEFAULT NOW()
);

CREATE INDEX idx_snapshots_contract_time ON option_chain_snapshots(contract_id, fetched_at DESC);

CREATE TABLE iv_history (
    symbol VARCHAR(10) NOT NULL,
    expiry DATE NOT NULL,
    atm_iv NUMERIC(8,6) NOT NULL,
    recorded_at TIMESTAMPTZ DEFAULT NOW(),
    PRIMARY KEY (symbol, expiry, recorded_at)
);
```

5.3 Redis Caching Strategy

CACHE KEY	TTL	CONTENT	RATIONALE
chain:{symbol}: {expiry}	55s	Full enriched chain JSON	Slightly less than 60s refresh to avoid serving stale data
quote:{symbol}	30s	Price, change, volume	Quotes change more frequently
expiries:{symbol}	300s	List of expiry dates	Expiry list is stable intraday
iv_surface:{symbol}	120s	Full surface grid	Expensive computation, less time-sensitive

All cache reads/writes are wrapped in try/except. If Redis is unavailable, `cache_get()` returns `None` (cache miss) and the endpoint fetches fresh data from yfinance. No errors propagate to the client.

Data Flow & Request Lifecycle

6.1 Options Chain Request

```

Browser: /chain page loads | └─ onMount() fires | └─ loadExpiries('SPY') → GET
/api/v1/options/expiries/SPY | └─ FastAPI → YFinanceProvider.get_expiries() [thread pool] |
└─ yfinance returns date strings → JSON response | └─ Frontend: expiries store set →
ExpirySelector renders | └─ $: if (browser && $selectedExpiry) loadChain() | └─
Promise.all([quote, chain]) in parallel | | | └─ GET /api/v1/market/quote/SPY | | └─
cache_get('quote:SPY') → miss | | └─ YFinanceProvider.get_quote() [thread pool] | | └─
yfinance.fast_info → QuoteData → cache_set → JSON | | | └─ GET /api/v1/options/chain/SPY?
expiry=...&strikes=30 | └─ cache_get('chain:SPY:2026-06-26') → miss | └─
provider.get_option_chain() [thread pool] | └─ yfinance.option_chain() → raw DataFrame | └─
.fi()/_ff() → NaN-safe parsing → OptionContractData list | └─ enrich_chain(): | for each
contract: | mid = (bid + ask) / 2 | iv = solve_iv(mid, S, K, T, r, q, type) | greeks =
calculate_greeks(S, K, T, r, q, iv, type) | sort by strike, mark ATM, compute stats | └─
cache_set('chain:SPY:...', result, ttl=55) | └─ ORJSONResponse → browser | └─ quote.set(q),
optionChain.set(chain) └─ OptionsChainTable re-renders └─ stat bar re-renders └─
startRefreshTimer(loadChain) → repeats every 60s

```

6.2 Browser Store Reactivity

```

User types 'NVDA' in SymbolSearch | └─ debounce 250ms | └─ GET /api/v1/symbols/search?q=NVDA →
results dropdown | └─ User clicks NVDA | └─ selectedSymbol.set('NVDA') | └─
chain/+page.svelte: $: if (browser && $selectedSymbol !== prevSymbol) | └─
selectedExpiry.set('') [reset expiry] | └─ loadExpiries('NVDA') [fetch new expiry list] | └─
Navbar: $selectedSymbol → input value updates (reactive) | └─ dashboard/+page.svelte: $: if
(browser && $selectedSymbol) load() | └─ re-fetches all analytics for NVDA | └─ All
components watching $selectedSymbol auto-update

```

Bugs Fixed & Engineering Decisions

This chapter documents every non-trivial bug encountered during development, its root cause, and the fix. Understanding these bugs builds deep knowledge of Python, JavaScript, Svelte 5, and Vite internals.

Bug 1 — `lib/` Gitignored

ASPECT	DETAIL
Symptom	All <code>frontend/src/lib/</code> files absent from git history
Root Cause	Python's default <code>.gitignore</code> includes <code>lib/</code> (for virtualenvs). Git silently ignored the entire frontend source directory.
Fix	Removed <code>lib/</code> line from <code>.gitignore</code>

Bug 2 — ECharts `.default` Does Not Exist

ASPECT	DETAIL
Symptom	Charts blank; TypeScript: <i>Property 'default' does not exist</i>
Root Cause	ECharts uses named exports, not a default export. <code>(await import('echarts')).default</code> is <code>undefined</code> , so <code>echarts.init()</code> throws <i>undefined is not a function</i> .
Wrong	<code>const echarts = (await import('echarts')).default;</code>
Fix	<code>const echarts = await import('echarts');</code>

Bug 3 — `onMount` Async Cleanup Ignored

ASPECT	DETAIL
Symptom	ResizeObserver / ECharts instance never cleaned up on component unmount; memory leak
Root Cause	Svelte's <code>onMount</code> expects either <code>() => void</code> (sync cleanup) or <code>Promise<never></code> . An async function returning a cleanup function produces <code>Promise<() => void></code> — Svelte ignores it entirely.
Fix	Non-async <code>onMount</code> launches an async IIFE internally, returns cleanup synchronously.

Bug 4 — `float('nan') or 0` Returns `nan`

ASPECT	DETAIL
Symptom	<code>GET /api/v1/options/chain/AAPL</code> → 500 Internal Server Error
Root Cause	<code>yfinance</code> returns <code>float('nan')</code> for missing volume/OI. In Python, <code>nan</code> is truthy, so <code>nan or 0</code> evaluates to <code>nan</code> , and <code>int(nan)</code> raises <code>ValueError: cannot convert float NaN to integer</code> .
Fix	Added <code>_fi()</code> / <code>_ff()</code> helpers using <code>math.isnan()</code> before casting.

```
# Why: bool(float('nan')) == True in Python
>>> float('nan') or 0
nan          # not 0 !
>>> int(float('nan'))
ValueError: cannot convert float NaN to integer
```

Bug 5 — SSR 500 on Every Page

ASPECT	DETAIL
Symptom	All pages return HTTP 500; browser shows <i>Internal Error</i>
Root Cause	SvelteKit SSR tried to render pages on the server. <code>fetch('/api/...')</code> with a relative URL fails server-side because there is no base URL — the Node.js <code>fetch</code> implementation requires a full URL.
Fix	<code>+layout.ts</code> : <code>export const ssr = false</code>

Bug 6 — Reactive Statements Run During SSR

ASPECT	DETAIL
Symptom	Pages still crashed during SSR even after adding <code>ssr = false</code> in layout
Root Cause	Svelte 5 executes <code>\$:</code> reactive declarations synchronously at component initialization — including during SSR. <code>onMount</code> doesn't run during SSR, but reactive declarations do. Fetch calls in reactive statements ran before <code>ssr = false</code> took effect.
Fix	Guard all reactive fetch calls: <code>\$: if (browser && \$selectedSymbol) load()</code>

Bug 7 — Blank Page (Root Cause: Wrong Svelte Build Served to Browser)

MOST COMPLEX BUG IN THIS PROJECT

This was the hardest bug to diagnose. The app silently produced a completely blank page with no visible errors. Required Chrome headless to capture the console error, then reading Svelte 5's internals to understand it.

ASPECT	DETAIL
Symptom	Completely blank page in browser. No HTML rendered. No visible error.
Console Error	Uncaught TypeError: Cannot read properties of null (reading 'r') at svelte.js:144
Root Cause	Svelte 5 ships two builds: <code>index-client.js</code> (browser) and <code>index-server.js</code> (server). The <code>package.json</code> exports use the <code>"browser"</code> condition to select the right one. SvelteKit's Vite plugin was not passing the <code>browser</code> condition to esbuild during dependency pre-bundling, so esbuild used the <code>"default"</code> export — which maps to <code>index-server.js</code> . The server build accesses <code>ssr_context.r</code> , but <code>ssr_context</code> is <code>null</code> in a browser environment. Every component crashed immediately. The bootstrap Promise chain swallowed the error silently.
Fix	Explicitly set <code>resolve.conditions</code> and <code>optimizeDeps.esbuildOptions.conditions</code> in <code>vite.config.ts</code> to <code>['browser', 'module', 'svelte', 'default']</code> .

```
// vite.config.ts – the fix
export default defineConfig({
  plugins: [sveltekit()],
  resolve: {
    conditions: ['browser', 'module', 'svelte', 'default'],
  },
  optimizeDeps: {
    include: ['svelte', 'echarts'],
    esbuildOptions: {
      conditions: ['browser', 'module', 'svelte', 'default'],
    },
  },
});
```

► Why `ssr_context.r` ?

In Svelte 5's server build (`index-server.js`), the `onDestroy` function reads:

```
function onDestroy(fn) {
  ssr_context.r.onDestroy(fn); // line 144 – ssr_context is null in browser!
}
```

`ssr_context` is a server-side global that tracks component lifecycle during HTML rendering. It is never initialized in a browser environment, so it is `null`, and accessing `null.r` throws the `TypeError`. The error is caught by the SvelteKit bootstrap `Promise.all().then()` but there is no `.catch()` handler, so the page silently stays blank.

Performance & Benchmarks

8.1 Quant Engine Performance

OPERATION	TARGET	ACHIEVED	METHOD
BS price (single)	< 1ms	~0.01ms	Pure Python + SciPy
BS price (10K contracts)	< 2,000ms	~4ms	NumPy vectorized
IV solve (Newton-Raphson)	< 10 iterations	3–5 iters	Brenner-Subrahmanyam initial guess
Full chain enrichment (30 strikes)	< 500ms	~80ms	Batch Greeks + IV solve
yfinance chain fetch	N/A	~400ms	Thread pool (non-blocking)

8.2 API Response Times (with Redis)

ENDPOINT	CACHE HIT	CACHE MISS
GET /options/chain	~5ms	~480ms
GET /market/quote	~2ms	~200ms
POST /pricing/black-scholes	N/A	~1ms
POST /strategy/payoff	N/A	~8ms
GET /iv/surface	~8ms	~900ms

8.3 Why NumPy is ~500x Faster

VECTORIZATION EXPLAINED

A Python for-loop prices one contract per iteration. Each iteration has Python interpreter overhead (~50ns per bytecode). For 10,000 contracts, that's millions of bytecodes.

NumPy replaces the Python loop with a single call to a C extension that operates on entire arrays. The C code runs at CPU speed with SIMD (Single Instruction Multiple Data) instructions — one CPU instruction processes 4–8 floats simultaneously. Result: the same computation runs ~500x faster.

8.4 Frontend Performance

METRIC	VALUE	NOTES
Initial page load (dev)	~600ms	Vite hot module replacement
ECharts canvas render (chain payoff)	~16ms	60fps-compatible
ECharts WebGL render (3D surface)	~8ms	GPU-accelerated
Store update → DOM re-render	< 1ms	Svelte's compiled output, no VDOM diffing

8.5 Why Svelte is Faster than React/Vue

React and Vue use a Virtual DOM (VDOM) — they maintain a JavaScript representation of the DOM, diff it on every update, and apply minimal patches. This diffing has non-zero overhead on every state change.

Svelte compiles components to vanilla JavaScript at build time. There is no VDOM, no diffing runtime, no framework overhead at runtime. A `$selectedSymbol = 'NVDA'` assignment compiles directly to a DOM update instruction. This is why Svelte re-renders are measured in microseconds.